Nirma University

Institute of Technology

Class Test, April 2026

M. Tech.(CSE/DS/CS), Semester-II

6CS252CC22: Research Methodology and IPR

Roll/ Exam No	25MCD005	Supervisor's initial with date	

Q-1 Ravi is an M.Tech student who has just completed his project on "Real-Time Object Detection using YOLOv8 on Edge Devices." He sits down to write his thesis and makes the following mistakes: **[10]**

- His abstract is 600 words long and written in future tense.
- His Literature Review is a list of 15 paper summaries, one after the other, with no grouping or identification of any research gap.
- In his Results section, he writes: *"The results were very good and much better than before."*
- He copies two paragraphs from a published paper and paraphrases one sentence without citing the source.
- His Methodology section describes only what he did, but does not justify why he chose YOLOv8 over other models.

(a) Identify all five mistakes Ravi has made and name the thesis section in which these mistakes were made.
(b) For any two of the identified mistakes, rewrite or correct that specific piece of content/text as it should ideally appear in the thesis.

Q-2 Dr Mehta is guiding three PhD students. Each has a different research problem: **[10]**

- **Student A – Priya:** Wants to study *why AI-based diagnostic tools are being resisted by doctors in rural hospitals*, by visiting two hospitals and conducting interviews.
- **Student B – Arjun:** Is developing a *new scheduling algorithm for cloud tasks* and wants to mathematically prove its time complexity is better than existing ones.
- **Student C – Sneha:** Has collected 5 years of *patient readmission records* and wants to build a model to predict high-risk patients.

Each student comes to Dr. Mehta saying, *"I will just start coding/implementing right away."*

(a)Identify the most appropriate investigation approach for each student (A, B, C) from the six broad categories discussed. Justify your choice in 2–3 sentences each.
(b)Dr. Mehta tells all three students their plan to "just start implementing" is a common research mistake. Name this mistake and explain two other common mistakes researchers make during investigation, with one example each.